

SIMATS Online Programming Lab

Java Programming — CO4-AT2-Laboratory Performance
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Q1. Menu Driven Calculator

Marks: 1 mark

Question: Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

Source Code

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import javax.imageio.ImageIO;
import java.io.File;

public class CalculatorApp extends JFrame implements ActionListener {

    JTextField num1Field, num2Field, resultField;
    JMenuBar menuBar;
    JMenu operationMenu;
    JMenuItem addItem, subItem, mulItem, divItem, clearItem, exitItem;

    public CalculatorApp() {
        setTitle("Kumaran S (192372048RB) - Menu Driven Calculator");
        setSize(420, 260);
        setLayout(new BorderLayout());

        // ---- Name banner on top ----
        JLabel nameLabel = new JLabel("Kumaran S - 192372048RB", SwingConstants.CENTER);
        nameLabel.setFont(new Font("SansSerif", Font.BOLD, 16));
        nameLabel.setOpaque(true);
        nameLabel.setBackground(new Color(25, 55, 110));
        nameLabel.setForeground(Color.WHITE);
        nameLabel.setBorder(BorderFactory.createEmptyBorder(8, 8, 8, 8));
        add(nameLabel, BorderLayout.NORTH);

        // ---- Menu Bar / Menu / MenuItem ----
        menuBar = new JMenuBar();
        operationMenu = new JMenu("Operations");

        addItem = new JMenuItem("Add");
        subItem = new JMenuItem("Subtract");
        mulItem = new JMenuItem("Multiply");
        divItem = new JMenuItem("Divide");
        clearItem = new JMenuItem("Clear");
        exitItem = new JMenuItem("Exit");

        addItem.addActionListener(this);
        subItem.addActionListener(this);
        mulItem.addActionListener(this);
        divItem.addActionListener(this);
        clearItem.addActionListener(this);
        exitItem.addActionListener(this);

        operationMenu.add(addItem);
        operationMenu.add(subItem);
        operationMenu.add(mulItem);
        operationMenu.add(divItem);
        operationMenu.addSeparator();
        operationMenu.add(clearItem);
        operationMenu.add(exitItem);

        menuBar.add(operationMenu);
        setJMenuBar(menuBar);

        // ---- Input panel ----
        JPanel panel = new JPanel(new GridLayout(3, 2, 8, 8));
```

```

        panel.setBorder(BorderFactory.createEmptyBorder(15, 15, 15, 15));
        panel.add(new JLabel("Number 1:"));
        num1Field = new JTextField();
        panel.add(num1Field);

        panel.add(new JLabel("Number 2:"));
        num2Field = new JTextField();
        panel.add(num2Field);

        panel.add(new JLabel("Result:"));
        resultField = new JTextField();
        resultField.setEditable(false);
        panel.add(resultField);

        add(panel, BorderLayout.CENTER);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        Object src = e.getSource();
        try {
            if (src == clearItem) {
                num1Field.setText("");
                num2Field.setText("");
                resultField.setText("");
                return;
            }
            if (src == exitItem) {
                System.exit(0);
            }

            double a = Double.parseDouble(num1Field.getText());
            double b = Double.parseDouble(num2Field.getText());
            double res = 0;

            if (src == addItem) res = a + b;
            else if (src == subItem) res = a - b;
            else if (src == mulItem) res = a * b;
            else if (src == divItem) {
                if (b == 0) {
                    resultField.setText("Error: Divide by 0");
                    return;
                }
                res = a / b;
            }
            resultField.setText(String.valueOf(res));
        } catch (NumberFormatException ex) {
            resultField.setText("Invalid Input");
        }
    }

    // Helper: auto-capture a screenshot of this window (used for demo/output capture only)
    private void captureScreenshot(String path) {
        try {
            Rectangle rect = this.getBounds();
            Robot robot = new Robot();
            Thread.sleep(400);
            java.awt.image.BufferedImage img = robot.createScreenCapture(rect);
            ImageIO.write(img, "png", new File(path));
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static void main(String[] args) throws Exception {
        CalculatorApp appFrame = new CalculatorApp();
        // Demo: pre-fill fields and trigger an addition, then screenshot
        SwingUtilities.invokeLater(() -> {
            appFrame.num1Field.setText("25");
            appFrame.num2Field.setText("15");
            appFrame.addItem.doClick();
        });
    }

```

```
Thread.sleep(800);  
appFrame.captureScreenshot("/home/claude/lab/screenshots/q1_output.png");  
System.exit(0);  
    }  
}
```

Output Screenshot

Operations	
Kumaran S - 192372048RB	
Number 1:	25
Number 2:	15
Result:	40.0